

Research Article

Observed and Near-Term CMIP6 Changes in Precipitation Extremes over Uttaradit Thailand

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Revised research draft requiring confirmation of data provenance before submission

Abstract

This study examines observed precipitation extremes at 13 stations in Uttaradit, Thailand, and near-term changes in seven available CMIP6 model series under SSP2-4.5 and SSP5-8.5. Eleven annual indices were calculated from observed 1981-2014 records and model historical (1995-2014) and future (2021-2050) series. Standard Mann-Kendall (MK) tests and Sen slopes provide the primary station-level trend description; the retained full-lag Yue-Wang modified MK is reported as a sensitivity analysis because its variance multiplier was below one for most short series. Future mean changes were evaluated model by model, then summarized as a multi-model median at each station followed by a median across stations; model spread and sign support are reported separately. Observed trends were heterogeneous: standard MK identified 30 significant station-index tests, with no province-wide field inference. The descriptive median PRCPTOT changes were 2.5% and 1.9% under SSP2-4.5 and SSP5-8.5, while median Rx1day changes were 6.6% and 8.6%. Heavy-threshold indices showed larger relative changes but should be interpreted with their low-frequency baselines and model spread. Future mean change is distinct from monotonic trend within 2021-2050. Station-based IDW is presented only as a visualization. Conclusions remain conditional on supplied data whose observation

provenance, bias-correction procedure, units and native calendar require documentary verification before submission.

Keywords: Precipitation extremes, Uttaradit, CMIP6, Mann-Kendall, Sen slope, Climate projections

1. Introduction

Annual rainfall totals alone cannot describe the intensity, timing and persistence of precipitation extremes. Standardized indices based on daily observations provide complementary information on heavy rainfall and dry or wet spells [1,2]. For engineering applications, distinguishing these measures helps frame questions about drainage, storage and drought exposure. An index change does not itself quantify infrastructure failure or flood discharge, which require additional hydrological and design information.

CMIP6 provides coordinated climate simulations for examining historical conditions and future scenarios [3,4]. Local interpretation nevertheless depends on model choice, spatial sampling and preprocessing of supplied series. Trend inference can also be sensitive to serial correlation treatment, especially for short annual records. The full-lag Yue-Wang effective-sample-size formulation [5] is retained as a sensitivity analysis, while conventional MK results and Sen slopes provide the primary station-level description.

The primary question is how observed precipitation extremes vary across Uttaradit and how the available CMIP6 models describe near-term changes under SSP2-4.5 and SSP5-8.5. Observed trends provide context, future mean change is the primary projection result, within-window future trend is a secondary diagnostic, and IDW is a station-based visualization only. The analysis distinguishes change in the 2021-2050 mean relative to 1995-2014 from monotonic trend within 2021-2050; these quantities need not have the same sign.

2. Materials and methods

2.1 Study network and supplied data

The network comprises stations 351001-351012 and 351201. Coordinates were read from the supplied station workbook. Observed records cover 1981-2014. Twenty-one supplied precipitation CSV files represent historical, SSP2-4.5 and SSP5-8.5 data from ACCESS-ESM1-5, CanESM5, CESM2, EC-Earth3, FGOALS-g3, MIROC6 and MRI-ESM2-0. The analysis uses the 13 target station columns. The filenames designate the model series as bias-corrected; the original correction algorithm, training period, validation and observational agency were not established from the files inspected. These details must be supplied and checked before submission. Rainfall values are treated as daily millimetres, an input assumption requiring confirmation from the data provider.

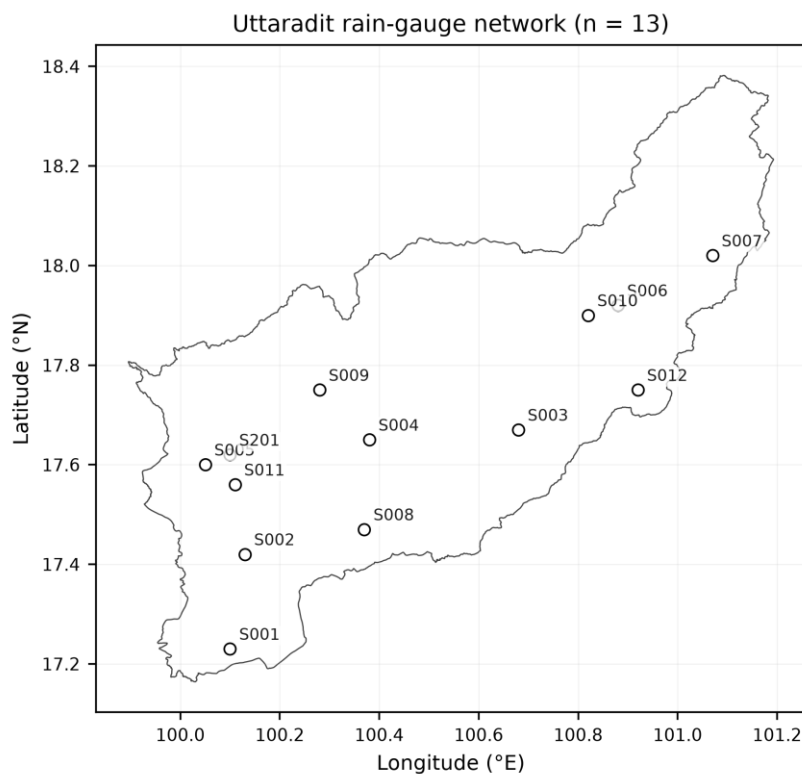


Figure 1 Locations of the 13 supplied stations and the Uttaradit polygon selected by province code 53 from the supplied GIS file.

2.2 *Quality control and annual indices*

Dates, station identifiers and coordinates were validated before calculation. Duplicate dates, negative rainfall and non-numeric rainfall are rejected. Each annual series is reindexed to a Gregorian calendar; a supplied record without February 29 therefore has that date missing, rather than having a synthetic value inserted. The 90% completeness denominator is 365 or 366 Gregorian days, as applicable. No missing day is imputed. An unrecorded day breaks a run and prevents a five-day accumulation spanning that gap. This is a study-specific annual completeness policy rather than a claim of full ETCCDI software equivalence. The native model calendar, rainfall variable, units and any upstream conversion must be confirmed from provider documentation before submission.

Wet days satisfy $RR \geq 1$ mm. PRCPTOT sums wet-day precipitation and SDII divides that sum by the number of wet days. Rx1day is the annual daily maximum; Rx5day is the maximum complete five-day running sum within the calendar year. CDD and CWD are the longest within-year runs below and at or above 1 mm, respectively. Spells and Rx5day windows are bounded by calendar years in this implementation. R10mm, R20mm and R50mm count days with rainfall at or above the stated threshold; R50mm is the selected user-defined threshold index. R95p and R99p sum rainfall strictly exceeding observed station-specific wet-day percentiles [1,2].

Percentiles use the full observed 1981-2014 reference sample and the NumPy linear quantile convention. The same thresholds are used for model historical and future series. No in-base-period bootstrap is applied; the resulting percentile indices are therefore described as fixed-reference ETCCDI-based indices. Threshold uncertainty is not propagated. The threshold CSV preserves the reference values and period.

2.3 Trend estimation

The Mann-Kendall statistic S sums the signs of all later-minus-earlier annual differences. Its variance includes the standard correction for tied values. A continuity-corrected normal statistic yields a two-sided p -value. Constant series return $S = 0$, $Z = 0$ and $p = 1$. At least ten valid annual values are required. Standard MK classifications and Sen slopes, the median of pairwise changes divided by actual differences in years [6], are the primary station-level results. A conventional 95% order-statistic slope interval is calculated with SciPy Theil-Sen routines; it is not adjusted for serial dependence.

As a prespecified sensitivity analysis, the retained Yue-Wang effective-sample-size procedure [5] removes the Sen trend before computing sample autocorrelations with a common mean and full-series sum-of-squares denominator. The variance multiplier is $f = 1 + 2 \sum[(1 - k/n) \rho(k)]$, for lags $k = 1, \dots, n - 1$; modified variance is $f \text{Var}(S)$, and effective sample size is n/f . All lags are included in the retained implementation. Factors below one are recorded explicitly because they can increase apparent significance. This short-series full-lag sensitivity procedure is not presented as superior to standard MK. Individual $\alpha = 0.05$ tests are unadjusted for multiplicity and do not establish regional field significance.

2.4 Future changes and spatial presentation

For each model, station and index, absolute change is the 2021-2050 annual mean minus the 1995-2014 historical annual mean. Relative change divides this difference by the historical mean and multiplies by 100; zero baselines do not produce finite percentages. Descriptive central values first take the multi-model median at each station, then the median across the 13 stations. Model spread is summarized from each model's station-median change using the interquartile range (IQR), and sign support is the number of the seven available models with positive or negative station-median change. These summaries are descriptive; the models are not independent realizations and no ensemble significance test is claimed.

Figure 5 distinguishes future mean change from within-window trend. Inverse-distance weighting with power two and great-circle distances is applied to station medians of absolute change [7]. Maps are masked to the explicitly identified Uttaradit polygon and displayed only over the calculated station-domain grid. Blank portions have no plotted estimate. Interpolation has not been cross-validated and is a station-based spatial visualization, not a validated gridded hazard estimate or a province-wide trend estimate.

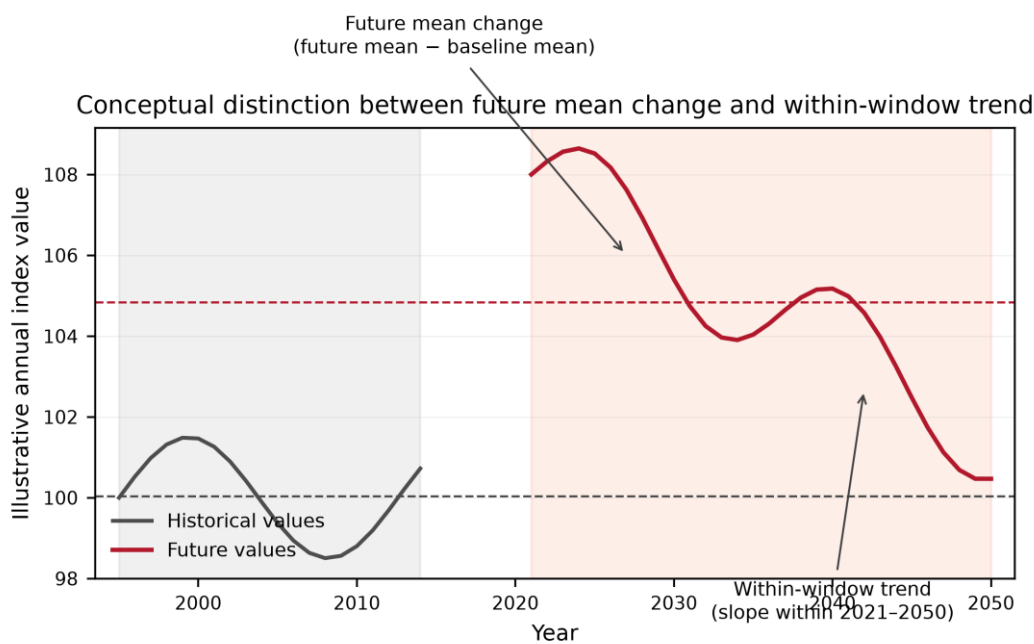


Figure 5 Conceptual schematic. Future mean change compares the 2021-2050 and 1995-2014 means, whereas within-window trend describes monotonic change inside 2021-2050. The lines are illustrative and are not Uttaradit observations or projections.

3. Results

3.1 Data coverage and observed trends

All 442 observed station-years and 5,460 future station-model-scenario-years met the implemented annual completeness criterion. This does not imply every day was present. The output contains 4,862 observed annual index values and 60,060 future values. Observed trend series have 34 annual values and future trend series have 30. The separate output checker

confirmed expected dimensions, bounded p-values, ordered slope intervals and model-consistent PRCPTOT means recomputed from raw daily input.

Observed standard-MK classifications were spatially mixed (Figure 2). Of 143 station-index tests, 30 were significant under standard MK. For PRCPTOT, no station increased, two decreased and 11 had no significant standard-MK trend. R50mm increased at two stations and decreased at one, whereas SDII decreased at four and increased at none. These counts refer to individual unadjusted tests; they do not demonstrate a province-wide trend after accounting for multiplicity or dependence. The full-lag Yue-Wang sensitivity analysis classified 66 tests as significant and is discussed as a method-sensitive secondary result.

Table 1 Observed standard-MK direction counts across the 13 stations, 1981-2014.

Index	Increasing	Decreasing	Not significant
PRCPTOT	0	0	13
SDII	2	2	9
Rx1day	3	1	9
Rx5day	1	1	11
CDD	1	3	9
CWD	2	2	9
R10mm	0	1	12
R20mm	2	1	10
R50mm	2	1	10
R95p	1	1	11
R99p	2	1	10

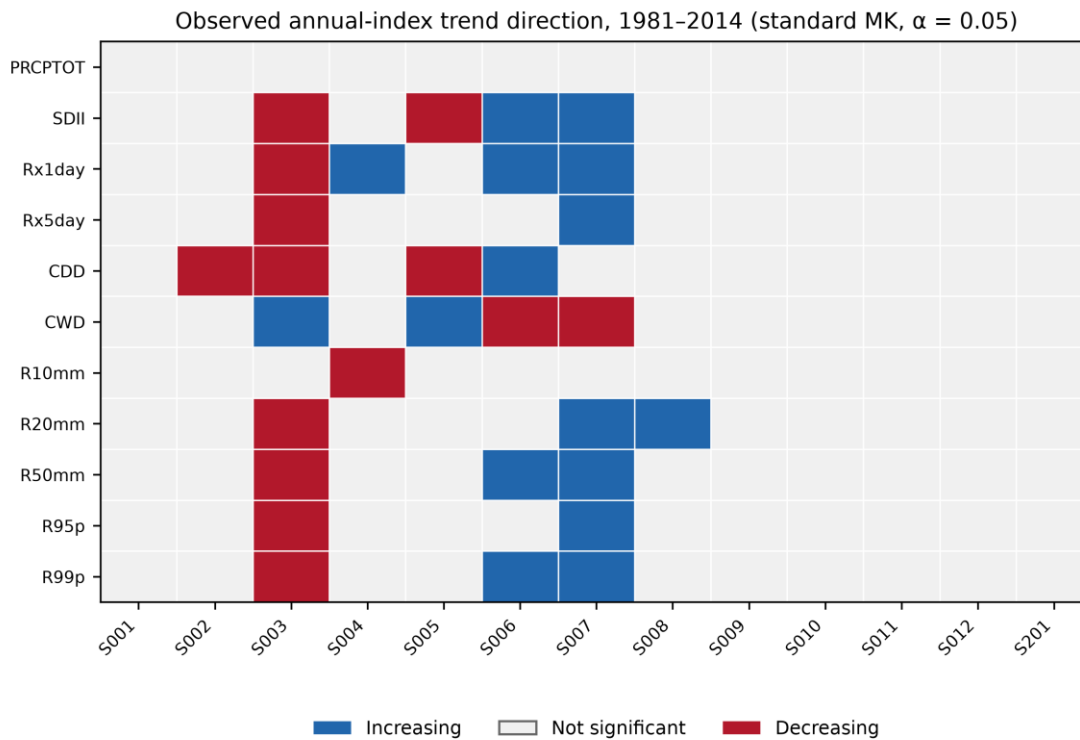


Figure 2 Individual observed station-index trend classifications from standard MK. S001-S012 and S201 denote the last three digits of the corresponding station IDs. Significance is unadjusted for multiple testing.

3.2 Projected mean changes and within-window trends

The station-first descriptive summaries indicate small positive PRCPTOT changes under both scenarios, while several heavy-rainfall measures show larger relative increases (Table 2; Figure 3). Rx1day rises by approximately 6.6% and 8.6%, and R50mm by 24.2% and 16.7%, under SSP2-4.5 and SSP5-8.5, respectively. R10mm and R20mm decrease slightly in both summaries. Table 2 reports the IQR and model sign support so that the central values are not interpreted as model consensus. Large percentage changes in infrequent exceedances need to be read with their absolute baseline frequencies.

Table 2 Multi-model summary of relative change (%) for 2021-2050 against each model's 1995-2014 baseline. Entries are central station-first median [IQR of seven model station-medians]; positive/negative model count.

Index	SSP2-4.5 median [IQR]; +/-	SSP5-8.5 median [IQR]; +/-
PRCPTOT	2.47 [-4.00, 4.85]; 4/3	1.93 [-5.03, 5.46]; 4/3
SDII	0.71 [-1.66, 3.41]; 4/3	-1.52 [-4.50, 3.35]; 3/4
Rx1day	6.62 [0.50, 15.38]; 7/0	8.63 [1.73, 13.81]; 5/2
Rx5day	4.27 [2.64, 15.61]; 6/1	3.70 [-1.99, 21.53]; 5/2
CDD	-0.49 [-3.66, -0.46]; 1/6	-0.97 [-3.52, 0.81]; 3/4
CWD	1.47 [-3.00, 8.17]; 4/3	2.65 [-0.33, 4.42]; 4/3
R10mm	-1.11 [-7.93, 3.33]; 3/4	-1.51 [-4.26, 3.00]; 3/4
R20mm	-2.29 [-7.89, 5.58]; 3/4	-2.29 [-11.25, 4.39]; 2/5
R50mm	24.24 [10.12, 29.31]; 6/1	16.67 [-5.30, 29.28]; 4/3
R95p	4.45 [0.58, 15.56]; 5/2	4.26 [-7.25, 19.48]; 4/3
R99p	29.22 [13.71, 33.47]; 7/0	25.09 [-5.54, 50.48]; 4/3

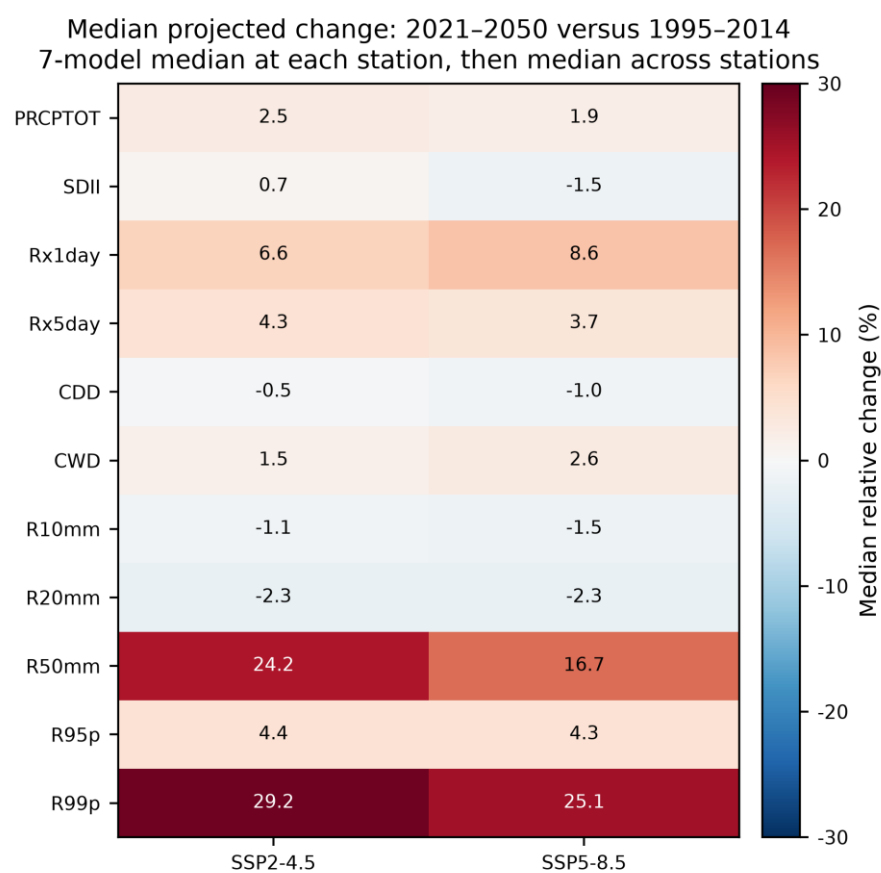


Figure 3 Descriptive changes with the same station-first aggregation as Table 2. Colours represent percentages, not significance.

Within the 2021-2050 window, full-lag Yue-Wang sensitivity classifications identified 445 of 1,001 tests under SSP2-4.5 and 390 of 1,001 under SSP5-8.5 as significant. Decreasing classifications outnumbered increasing classifications in both scenarios (260 versus 185 and 223 versus 167). These within-window sensitivity results can coexist with a future mean above the earlier baseline because they measure different quantities (Figure 5). Figure 4 illustrates station-median PRCPTOT changes and spatial differences between scenarios without assigning significance to interpolated cells.

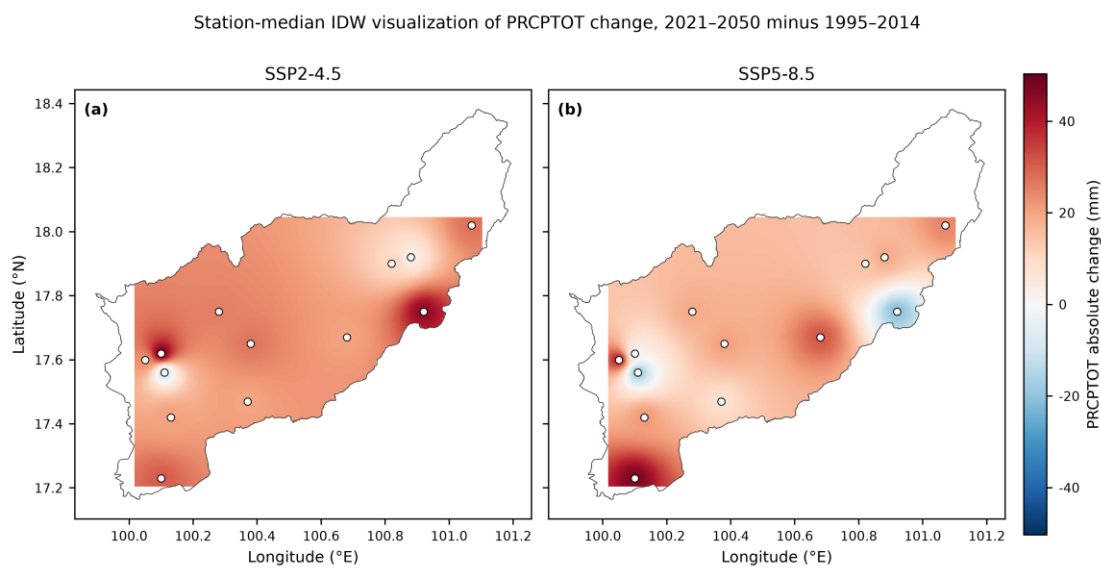


Figure 4 IDW of seven-model station medians of absolute PRCPTOT change. Both panels share a colour scale. White areas outside the displayed station-domain estimate are not zero change. The polygon mask is Uttaradit only.

4. Discussion

Observed extremes are heterogeneous across the station network. The standard-MK results do not support a single province-wide direction for PRCPTOT or most other indices, and the unadjusted classifications must not be interpreted as field significance. The descriptive projection pattern separates rainfall accumulation from heavy-event metrics: small median PRCPTOT changes coexist with larger central changes for Rx1day and selected threshold indices. This can motivate separate assessment of water supply and intense rainfall exposure,

but it cannot determine drainage capacity, return-period rainfall or flood damage from these indices alone. Likewise, CDD is a precipitation-spell measure and does not incorporate evapotranspiration or soil water storage.

Model choice is a material source of uncertainty. The multi-model median is a descriptive central value, not evidence that the seven models are independent or agree in sign. The IQR and positive/negative model counts in Table 2 should accompany any interpretation of the median. Large relative changes in R50mm, R95p and R99p require particular care because percentage changes can be amplified when the historical frequency or baseline amount is low. Future mean change is also not a within-window trend: a future period can have a higher mean than the historical baseline while showing a decreasing monotonic tendency within that future window.

The discrepancy between standard and modified test classifications is a substantive inference sensitivity. The full-lag multiplier falls below one for 128 of 143 observed series and 1,992 of 2,002 future series; consequently this sensitivity analysis often becomes more permissive. It is not described as automatically more conservative or more reliable. A purpose-built simulation study covering sample sizes of 30 and 34 years and relevant autocorrelation structures would be required before a different modified trend procedure could be adopted as primary. The reported conventional Sen intervals retain their independence assumptions.

Additional limits arise upstream. Before submission, data-provider documentation must establish the observational agency and quality control, rainfall variable and units, native model calendars and any conversion, spatial sampling, bias-correction method, reference observations, calibration period and validation evidence. A filename alone does not verify preprocessing performance. Fixed observed percentiles aid comparability to a common historical threshold but omit threshold-estimation uncertainty and differ from bootstrap treatments used in some ETCCDI implementations. The analysis is therefore a reproducible,

model-conditioned assessment of the supplied series, not a validated engineering design standard or deterministic province-wide forecast.

5. Conclusions

Observed precipitation extremes are heterogeneous across the 13 stations; standard-MK station tests do not establish a province-wide trend. Descriptive near-term multi-model median changes are modest for PRCPTOT but larger for selected heavy-rainfall indices, including Rx1day and R50mm. These changes should be interpreted jointly with model spread, sign support and the absolute baselines of infrequent indices. Future mean change and within-window trend answer different questions and may have different signs. Finally, statistical classifications are sensitive to the selected trend-test treatment, and all projection inferences remain model-conditioned until data provenance, bias correction, units and calendar treatment are documented and verified.

6. Acknowledgements and declarations

[To be completed by the authors: data-provider acknowledgement, funding, conflicts of interest, author contributions and any required AI-use disclosure. No declarations have been inferred.]

Data and code availability: The revised code, regression tests, input-file manifest, QC tables and result workbook accompany this draft locally. [Authors must supply data-use permissions and a public repository or justified access statement.]

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